Ruhan Bhavsar

365-998-6224 | r5bhavsa@uwaterloo.ca | Linkedin | Github

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechatronics Engineering — *Sept 2024 - Present*

Experience

Engineering Intern — Jan 2026 - Apr 2026
MPI Morheat — *Toronto, ON*

- Designed and fabricated an aluminum plate bending roller mechanism using **welded components** and **geared rollers**, improving heater plate forming consistency and reducing manual forming time from **20 minutes to 5 minutes per plate**.
- Developed a low-cost automated **three-axis label cutter** for under **$500** using **stepper motors**, **3D-printed components**, and an **STM32 microcontroller** programmed in **C++**, saving **5–10 hours** of manual labor per week.
- Developed a **React-based** configurator for flange, screwplug, and PTFE heater drawings, reducing custom drawing generation time by **95%** from approximately **20 minutes to 1 minute** for customer proposals.

Engineering Intern — May 2025 - Aug 2025
Vexos — *Markham, ON*

- Redesigned **SMT rework nozzle** in **SolidWorks** and **SolidWorks PDM** for local machining, reducing supplier lead time from **6 weeks to 1 week** and lowering per-unit cost by $\sim$ **$700.**
- Designed and **3D printed** conformal coating caps in **SolidWorks** to protect PCB components during coating, improving process repeatability and supporting higher quality-control pass rates.
- Engineered an **automated curing stand** for **optical lens assemblies**, eliminating **1.5 minutes** of operator supervision per board and saving several hours weekly.

Firmware Developer — Sept 2024 - Present
Midnight Sun Solar Car Team — *Waterloo, ON*

- Maintained embedded firmware for **8 vehicle microcontrollers**, managing **ADC sampling** and **real-time tasks** with a **10 ms latency target** for system synchronization.
- Built a **Pytest-based Hardware-in-the-Loop testing framework** to validate firmware directly on **vehicle hardware**, reducing manual testing time by **80%**.
- Programmed and tested embedded systems in **C and Python** using **SCons**, **Vagrant**, and **Git-based development workflows**.

Projects

Robotic Hand | *Python, MediaPipe, ESP32*

- Designed and **3D printed** a **six-finger robotic hand** with **servo-driven actuation** for real-time motion replication.
- Developed a **Python-based** computer vision system using **MediaPipe** to detect finger states (open, closed, and intermediate).
- Implemented real-time serial communication between **Python** and **ESP32 firmware** to translate detected hand positions into **servo commands**.

Rescue Robot | *C++, RobotC, AutoCAD, SolidWorks*

- Designed and modeled **robot chassis and mechanical components** in **SolidWorks** and **AutoCAD** for autonomous rescue-supply delivery.
- Programmed a **Lego EV3** robot in **RobotC** to navigate a grid using a Depth-First Search pathfinding algorithm.
- Integrated autonomous navigation with **pickup and drop-off mechanisms** to complete object delivery tasks.

Technical Skills

Mechanical Design & Manufacturing: SolidWorks, SolidWorks PDM, AutoCAD, 3D Printing, Bambu Studio, fixture design, prototyping, technical drawings
Embedded & Electrical: STM32, ESP32, Arduino, ADCs, RTOS concepts, servo motors, stepper motors, PySerial
Programming: C, C++, Python, MATLAB, JavaScript, TypeScript, SQL, HTML/CSS
Tools & Workflow: Git, Linux, Excel, SCons, Vagrant, Asset Essentials